

Sensors & Systems

Exercises — Lecture 2

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Exercise 1 Valid PDF, Expectation and Variance

Problem statement

Is the function

$$f(x) = \begin{cases} \frac{3}{5}x & \text{if } 0 < x \leq 1, \\ \frac{(\sqrt{3}x - \sqrt{12})^2}{5} & \text{if } 1 < x \leq 3, \\ \frac{12}{5} - \frac{3}{5}x & \text{if } 3 < x \leq 4, \\ 0 & \text{otherwise,} \end{cases}$$

a valid probability density function? If yes, find the expectation μ_X and variance σ_X^2 of the associated random variable.

Simulation: [exercise_1.py](#) — click to open the live-sampling script (works in most PDF viewers).

Solution

What does $f(x)$ actually tell us?

$f(x)$ is a *density* — it does not give the probability at a single point. Probability only comes from integrating over an interval:

$$P(a \leq X \leq b) = \int_a^b f(x) dx \quad (\text{area under the curve})$$

The three pieces simply describe the *shape* of the density over different regions of x :

Piece	Range	Formula	Shape / meaning
1	$0 < x \leq 1$	$\frac{3}{5}x$	Ramp up — values near $x = 1$ more likely than near $x = 0$
2	$1 < x \leq 3$	$\frac{3}{5}(x - 2)^2$	Parabola — $x = 2$ is the least likely point in this piece
3	$3 < x \leq 4$	$\frac{12}{5} - \frac{3}{5}x$	Ramp down — values near $x = 3$ more likely than near $x = 4$
—	otherwise	0	Impossible — X cannot take values outside $[0, 4]$

For example, $P(0 \leq X \leq 1) = I_1 = \frac{3}{10} = 30\%$.

Part 0 — Simplify the middle piece

First note that the middle piece simplifies:

$$\frac{(\sqrt{3}x - \sqrt{12})^2}{5} = \frac{3x^2 - 2\sqrt{36}x + 12}{5} = \frac{3x^2 - 12x + 12}{5} = \frac{3(x-2)^2}{5}$$

So we work with:

$$f(x) = \begin{cases} \frac{3}{5}x & 0 < x \leq 1 \\ \frac{3}{5}(x-2)^2 & 1 < x \leq 3 \\ \frac{12}{5} - \frac{3}{5}x & 3 < x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

Part 1 — Is $f(x)$ a valid PDF?

A function is a valid PDF if (1) $f(x) \geq 0$ for all x , and (2) $\int_{-\infty}^{\infty} f(x) dx = 1$.

Non-negativity: each piece is non-negative on its domain (a linear ramp, a squared term, and a falling ramp respectively). ✓

Integrates to 1:

The following integration rules are used throughout:

Rule	Form	Result
R1	$\int x^n dx$	$\frac{x^{n+1}}{n+1} + C$ (power rule)
R2	$\int (x-a)^n dx$	$\frac{(x-a)^{n+1}}{n+1} + C$ (shifted power rule)
R3	$\int c dx$	$cx + C$ (constant rule)

Piece 1 $I_1 = \int_0^1 \frac{3}{5}x dx$

Using **R1** with $n = 1$: $\int x dx = \frac{x^2}{2}$, so

$$I_1 = \frac{3}{5} \cdot \frac{x^2}{2} \Big|_0^1 = \frac{3}{10} [x^2]_0^1 = \frac{3}{10} [(1)^2 - (0)^2] = \frac{3}{10} \cdot 1 = \frac{3}{10}$$

Piece 2 $I_2 = \int_1^3 \frac{3}{5}(x-2)^2 dx$

Using **R2** with $a = 2$, $n = 2$: $\int (x-2)^2 dx = \frac{(x-2)^3}{3}$, so

$$I_2 = \frac{3}{5} \cdot \frac{(x-2)^3}{3} \Big|_1^3 = \frac{1}{5} [(x-2)^3]_1^3 = \frac{1}{5} [(3-2)^3 - (1-2)^3] = \frac{1}{5} [(1)^3 - (-1)^3] = \frac{1}{5} (1+1) = \frac{2}{5}$$

Piece 3 $I_3 = \int_3^4 \left(\frac{12}{5} - \frac{3}{5}x \right) dx$

Split into two terms. Using **R3** on the constant and **R1** ($n = 1$) on x :

$$\int \left(\frac{12}{5} - \frac{3}{5}x \right) dx = \frac{12}{5}x - \frac{3}{5} \cdot \frac{x^2}{2} = \frac{12x}{5} - \frac{3x^2}{10}$$

Inserting the limits:

$$\begin{aligned} I_3 &= \left[\frac{12x}{5} - \frac{3x^2}{10} \right]_3^4 = \left(\frac{12 \cdot 4}{5} - \frac{3 \cdot 16}{10} \right) - \left(\frac{12 \cdot 3}{5} - \frac{3 \cdot 9}{10} \right) \\ &= \left(\frac{48}{5} - \frac{48}{10} \right) - \left(\frac{36}{5} - \frac{27}{10} \right) = \left(\frac{96}{10} - \frac{48}{10} \right) - \left(\frac{72}{10} - \frac{27}{10} \right) = \frac{48}{10} - \frac{45}{10} = \frac{3}{10} \end{aligned}$$

Total:

$$I_1 + I_2 + I_3 = \frac{3}{10} + \frac{4}{10} + \frac{3}{10} = \frac{10}{10} = 1 \quad \checkmark$$

Yes, $f(x)$ is a valid PDF.

Part 2 — Expectation $\mu_X = E[X]$

The **expectation** is the probability-weighted average of x :

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

Each piece contributes $\int x \cdot f(x) dx$ over its own interval. The same rules R1–R3 from Part 1 apply; for Pieces 2 and 3 the product $x \cdot f(x)$ is expanded into a polynomial first, then each term is integrated with R1.

$$E[X] = \underbrace{\int_0^1 x \cdot \frac{3x}{5} dx}_{J_1} + \underbrace{\int_1^3 x \cdot \frac{3(x-2)^2}{5} dx}_{J_2} + \underbrace{\int_3^4 x \left(\frac{12}{5} - \frac{3x}{5} \right) dx}_{J_3}$$

Piece 1 $J_1 = \int_0^1 \frac{3x^2}{5} dx$

Multiply: $x \cdot \frac{3x}{5} = \frac{3x^2}{5}$. Using **R1** with $n = 2$: $\int x^2 dx = \frac{x^3}{3}$, so

$$J_1 = \frac{3}{5} \cdot \frac{x^3}{3} \bigg|_0^1 = \frac{1}{5} [x^3]_0^1 = \frac{1}{5} [(1)^3 - (0)^3] = \frac{1}{5}$$

Piece 2 $J_2 = \int_1^3 \frac{3}{5} x(x-2)^2 dx$

Expand $x(x-2)^2 = x(x^2 - 4x + 4) = x^3 - 4x^2 + 4x$, then apply **R1** to each term:

$$\int (x^3 - 4x^2 + 4x) dx = \frac{x^4}{4} - \frac{4x^3}{3} + 2x^2$$

Inserting the limits:

$$\begin{aligned} J_2 &= \frac{3}{5} \left[\frac{x^4}{4} - \frac{4x^3}{3} + 2x^2 \right]_1^3 = \frac{3}{5} \left[\left(\frac{81}{4} - \frac{108}{3} + 18 \right) - \left(\frac{1}{4} - \frac{4}{3} + 2 \right) \right] \\ &= \frac{3}{5} \left[\left(\frac{81}{4} - 36 + 18 \right) - \left(\frac{3}{12} - \frac{16}{12} + \frac{24}{12} \right) \right] = \frac{3}{5} \left[\frac{9}{4} - \frac{11}{12} \right] = \frac{3}{5} \cdot \frac{27-11}{12} = \frac{3}{5} \cdot \frac{4}{3} = \frac{4}{5} \end{aligned}$$

Piece 3 $J_3 = \int_3^4 x \left(\frac{12}{5} - \frac{3x}{5} \right) dx$

Multiply out: $x \left(\frac{12}{5} - \frac{3x}{5} \right) = \frac{1}{5}(12x - 3x^2)$. Apply **R1** to each term:

$$\int (12x - 3x^2) dx = 6x^2 - x^3$$

Inserting the limits:

$$J_3 = \frac{1}{5} [6x^2 - x^3]_3^4 = \frac{1}{5} [(6 \cdot 16 - 64) - (6 \cdot 9 - 27)] = \frac{1}{5} [(96 - 64) - (54 - 27)] = \frac{1}{5} [32 - 27] = 1$$

$$\mu_X = J_1 + J_2 + J_3 = \frac{1}{5} + \frac{4}{5} + 1 = 2$$

Part 3 — Variance $\sigma_X^2 = E[X^2] - \mu_X^2$

The **variance** measures how spread out X is around its mean. The shortcut formula avoids computing $E[(X - \mu)^2]$ directly:

$$\sigma_X^2 = E[X^2] - (E[X])^2$$

We compute $E[X^2] = \int x^2 f(x) dx$ (same structure as before, multiply x^2 into each piece), then subtract $\mu_X^2 = 2^2 = 4$.

$$E[X^2] = \underbrace{\int_0^1 x^2 \cdot \frac{3x}{5} dx}_{K_1} + \underbrace{\int_1^3 x^2 \cdot \frac{3(x-2)^2}{5} dx}_{K_2} + \underbrace{\int_3^4 x^2 \left(\frac{12}{5} - \frac{3x}{5} \right) dx}_{K_3}$$

Piece 1 $K_1 = \int_0^1 \frac{3x^3}{5} dx$

Multiply: $x^2 \cdot \frac{3x}{5} = \frac{3x^3}{5}$. Using **R1** with $n = 3$: $\int x^3 dx = \frac{x^4}{4}$, so

$$K_1 = \frac{3}{5} \cdot \frac{x^4}{4} \Big|_0^1 = \frac{3}{20} [x^4]_0^1 = \frac{3}{20} [(1)^4 - (0)^4] = \frac{3}{20}$$

Piece 2 $K_2 = \int_1^3 \frac{3}{5} x^2 (x-2)^2 dx$

Expand $x^2(x-2)^2 = x^2(x^2 - 4x + 4) = x^4 - 4x^3 + 4x^2$, then apply **R1** to each term:

$$\int (x^4 - 4x^3 + 4x^2) dx = \frac{x^5}{5} - x^4 + \frac{4x^3}{3}$$

Inserting the limits:

$$\begin{aligned} K_2 &= \frac{3}{5} \left[\frac{x^5}{5} - x^4 + \frac{4x^3}{3} \right]_1^3 = \frac{3}{5} \left[\left(\frac{243}{5} - 81 + \frac{108}{3} \right) - \left(\frac{1}{5} - 1 + \frac{4}{3} \right) \right] \\ &= \frac{3}{5} \left[\left(\frac{243}{5} - 81 + 36 \right) - \left(\frac{3}{15} - \frac{15}{15} + \frac{20}{15} \right) \right] = \frac{3}{5} \left[\frac{18}{5} - \frac{8}{15} \right] = \frac{3}{5} \cdot \frac{54 - 8}{15} = \frac{3}{5} \cdot \frac{46}{15} = \frac{46}{25} \end{aligned}$$

Piece 3 $K_3 = \int_3^4 x^2 \left(\frac{12}{5} - \frac{3x}{5} \right) dx$

Multiply out: $x^2(\frac{12}{5} - \frac{3x}{5}) = \frac{1}{5}(12x^2 - 3x^3)$. Apply **R1** to each term:

$$\int (12x^2 - 3x^3) dx = 4x^3 - \frac{3x^4}{4}$$

Inserting the limits:

$$\begin{aligned} K_3 &= \frac{1}{5} \left[4x^3 - \frac{3x^4}{4} \right]_3^4 = \frac{1}{5} \left[\left(4 \cdot 64 - \frac{3 \cdot 256}{4} \right) - \left(4 \cdot 27 - \frac{3 \cdot 81}{4} \right) \right] \\ &= \frac{1}{5} \left[(256 - 192) - \left(108 - \frac{243}{4} \right) \right] = \frac{1}{5} \left[64 - \frac{189}{4} \right] = \frac{1}{5} \cdot \frac{256 - 189}{4} = \frac{1}{5} \cdot \frac{67}{4} = \frac{67}{20} \end{aligned}$$

Total $E[X^2]$: convert to a common denominator of 100:

$$E[X^2] = \frac{3}{20} + \frac{46}{25} + \frac{67}{20} = \frac{15}{100} + \frac{184}{100} + \frac{335}{100} = \frac{534}{100} = \frac{267}{50}$$

$$\sigma_X^2 = E[X^2] - \mu_X^2 = \frac{267}{50} - 4 = \frac{267 - 200}{50} = \frac{67}{50}$$

Answers: $\mu_X = 2$, $\sigma_X^2 = \frac{67}{50} = 1.34$

Part 4 — Python Simulation

Script: `exercise_1.py`

Samples the random variable using **inverse-CDF (inverse transform) sampling**:

draw $u \sim \mathcal{U}(0, 1)$ and apply $F^{-1}(u)$, where the CDF pieces are:

$$F(x) = \begin{cases} \frac{3x^2}{10} & 0 < x \leq 1 \\ \frac{3}{10} + \frac{(x-2)^3 + 1}{5} & 1 < x \leq 3 \\ \frac{24x - 3x^2 - 38}{10} & 3 < x \leq 4 \end{cases}$$

Inverting each piece gives:

$$F^{-1}(u) = \begin{cases} \sqrt{10u/3} & u \leq 3/10 \\ 2 + \sqrt[3]{5u - 5/2} & 3/10 < u \leq 7/10 \\ 4 - \sqrt{120(1-u)}/6 & u > 7/10 \end{cases}$$

The live plot shows the histogram converging to $f(x)$, the true expectation $\mu = 2$ as a dashed line, and running sample mean and variance converging to their true values.